

Investigating the impact of Open Spaces on Quality of Life at Zonal level for Bangalore city through Geospatial techniques

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Abstract— This study explores the impact of urban open spaces on improving the quality of life in rapidly growing cities, with a particular focus on Bengaluru. It emphasizes the ecological, social, and psychological advantages of these areas, such as temperature, biodiversity preservation, and stress alleviation. Despite these advantages, Bengaluru encounters issues like insufficient open space, comprising only 3% of the city's area, restricted accessibility, and inadequate maintenance of current parks. Utilizing remote sensing techniques, such as Landsat 8 satellite imagery from 2015 to 2025, the study assesses changes in vegetation cover, surface temperatures, and built-up areas utilizing indices such as NDVI, SI, VCI, LST, NDBI, and NDWI. The results indicate a reduction in vegetation and water bodies alongside an increase in developed areas, highlighting the effects of urbanization. To improve Bengaluru's liveability and quality of life through the thoughtful creation of open spaces, the report recommends smart urban planning and the efficient use of remote sensing.

Keywords—Open Spaces, Quality of Life, Remote Sensing, Urban Green Space, Geographic Information System (GIS)

I. INTRODUCTION

Maintaining the vibrancy of a city has made open spaces one of the most crucial parts of city. Indian cities have seen a decline in both the number and quality of open spaces over time, and this has to be addressed immediately. The idea of "open space" has changed dramatically throughout time, reflecting its significance in environmental preservation and urban development. In 1833, at a "public trial in London," a committee used the word "open space" for the first time. The "Metropolitan Open Space Act," which was passed in London, Britain, in 1877 [2]. Open areas help in

1. Temperature Regulation: Heat and noise pollution are lessened by urban greenery.
2. Conservation of Biodiversity: Open spaces protect habitats and ecosystems.
3. Stress Reduction: They provide a space for rest and leisure.
4. Community Identity: Encourage a feeling of togetherness and belonging.
5. Youth Involvement: Community initiatives foster strong relationships and self-worth.
6. Social Harmony: Promote communication between various ethnic groups to improve cross-cultural understanding.

There are different types of open spaces which are categorized as Traditional and Innovative open spaces. Traditional consists of public, neighbourhood parks, pedestrian malls, plazas, etc. while innovative consists of community and neighbourhood open spaces, schoolyards, streets, etc. [1]. Quality of life is a complex phenomenon composed of basic elements influenced by both personal and

environmental factors. Quality of life is a complicated concept with many significant facets that are influenced by both individual and environmental factors. It encompasses social, emotional, and physical well-being and can be influenced by things like neighbourhood conditions, job, education, housing, and transportation [17].

The quality of life is shaped by how people interact with their urban surroundings. By providing for the demands of recreation, health, and a comfortable atmosphere, open spaces enhance people's quality of life. Gardens and trees are examples of features that might promote stress relief and relaxation. Open spaces and quality of living have a complex relationship. Numerous benefits are linked to having access to parks and recreational areas. Research has demonstrated that being in a natural environment can help people relax and feel less depressed and anxious. As a physical structure and a venue for a variety of activities, public open space significantly improves people's quality of life, particularly by meeting their requirements for recreation, health, and a nice urban setting. India's urban population has grown over fivefold in the last fifty years, and the country now contains twenty-two megacities, up from four in 1975 [10].

Open spaces have always been used in Indian towns, even though they are essential for leisure, conservation, and urban beauty. Global criteria are determined by local factors, even though the World Health Organization recommends that every resident have at least 9 square meters of green open space [9]. Developed countries, on the other hand, strive for greater green space standards (20–40) square meters per capita) to enhance ecological balance and human well-being [1]. Urban open spaces which include green spaces like parks.

Hierarchy of the Urban open spaces

Parks at the regional level are big parks that are 20–40 acres in size. These serve as important leisure centres for larger areas. Tot Lots are small areas up to 150 m² that are frequently used by very young children and their families. Neighbourhood-Level Parks: These are moderately sized parks, ranging from 150 to 400 m². City-Level Parks: These parks, which range in size from 1 to 20 acres, are usually the site of bigger meetings and city-wide events. State-Level Parks: Covering 40–100 acres, these important sites include, for example, 60-acre Brindavan Gardens. National-level parks are large, ecologically and recreationally significant places that are larger than 100 acres, like Bannerghatta National Park or the Sundarbans. [9]

In Bangalore city the total number of parks at present are 1115 parks and 192 playgrounds. Out of which 119 parks are present in Bommanahalli zone, 54 are present in Dasarahalli

zone, 269 in East zone, 48 in Mahadevapura zone, 139 in RR.Nagara, 285 in South zone, 220 in West zone and 102 in Yelahanka zone.

Remote sensing offers insights into impervious surfaces, surface temperatures, land cover changes, and urban sprawl by substituting quick, extensive data collecting for labour-intensive, sluggish human surveys. Green space distribution, air quality, and microclimates can now be thoroughly evaluated in days rather than years form the techniques of remote sensing through NDVI (vegetation health), and thermal imaging (urban heat islands), vegetation condition index, shadow index. It will monitor and identify pollution levels close to parks, establishing a connection between green space and health outcomes. Real-time monitoring and data-driven urban policies are made possible, guaranteeing prompt actions to improve liveability and encourage sustainable urban development. The aim of the study is to evaluate the role of urban open spaces in enhancing the quality of life, by utilizing remote sensing techniques and GIS tools.

II. INTRODUCTION TO THE CITY

A. About the City

At an elevation of around 931 meters above mean sea level, Bangalore is located in the center of the South Indian Peninsula, evenly distanced from the eastern and western coasts (latitude 12° 58 North and longitude 73° 36 East). It is well known for having a healthy and balanced atmosphere [10]. Bengaluru, the capital of Karnataka, is India's fifth-largest metropolitan area. Established in 1537, it grew from a tiny community to become India's preeminent scientific and technological hub and a fierce rival on the world stage. Between 1985 and 2015, the city's built-up area increased from 192 to 518 square kilometers, a 170% increase. The Bruhat Bengaluru Mahanagara Palike (BBMP), which spans over 713 square kilometers, was created in 2007 by the merger of the Bangalore City Corporation and the neighboring territories. According to the 2011 Census, the 713 square kilometer municipal corporation was home to 84, 43,675 (8.43 million) people. [10].

Fig 1 represents the location of the study area and its location in India. Fig 2 shows the population map of the city (2011 census) and population density distribution by ward, and it is evident that wards in the central city have higher densities than wards outside of it. Unplanned urban expansion resulted from the city having to grow outside of its core to meet the growing push for growth.

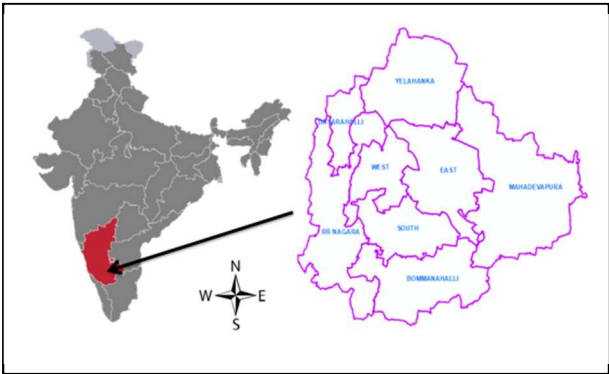


Fig 1: Study Area of Bangalore City

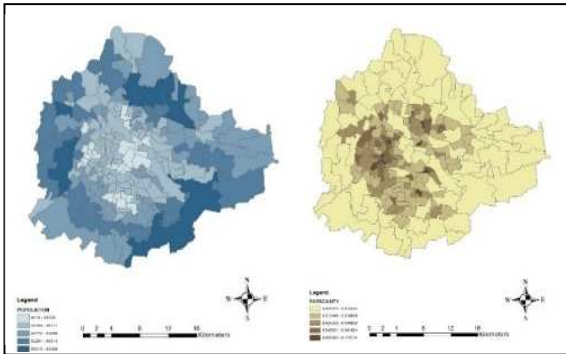


Fig 2: Population and Population Density Map respectively

B. Condition of the city related to Open Spaces

Bangalore, once celebrated as the "Garden City of India. Parks like Cubbon Park and Lalbagh Garden are crucial for Bangalore's green city image, providing much-needed recreational spaces, preserving nature, and serving as cultural landmarks. They cater to a variety of needs: relaxation, engagement, cognitive, and aesthetic. They hold symbolic meaning for citizens who have formed connections with these spaces over generations. In 1983, Bangalore's parks, playgrounds, and structured open space covered 2050 hectares, or 10.1% of the city's total area, according to the city's development plan (CDP). In contrast, open space made for 17.2% of the entire area in 1960. One of the metropolises with the quickest rate of growth is Bangalore [9]. Bangalore's urbanized area has been growing steadily and widely over time. About 8.4% of the city's land is set aside for green space, which is not enough for the more than 12 million people who call it home. This percentage is much lower than the international norms that health groups advise. The usability of the surviving parks for leisure and relaxation is restricted by their frequent overcrowding and inadequate maintenance.

With an area of 709 square kilometers and a population of 12 million (an estimated estimate of 8.4 million according to the 2011 census), Bengaluru only has 19.31 square kilometers set aside for open spaces like parks and playgrounds. This translates to 2.2 square feet of open space per person, which is much less than the 10–12 square feet that the Ministry of Urban Development, India's URDPFI (Urban and Regional Development Plans Formulation and Implementation) guidelines prescribe.

III. DATA AND METHODS

This study uses a mixed-method approach to assess Bangalore's open spaces and their effect on quality of life. Landsat 8 imagery (2015–2025) and spectral indices (NDVI, NDBI, NDWI, VCI, SI, LST) are analysed remotely to monitor changes in vegetation, water bodies, surface temperatures, and built-up regions. In order to evaluate open space distribution, population density, and park conditions across zones and wards, this data is combined with 2011 Census and BBMP data. And at the end all the indices and the secondary data were combined and results and discussions were obtained. Fig 3 shows the detail methodology flow chart. Table 1 and 2 shows the steps and formulas and their purpose used to achieve the

results of various indices used in the study.

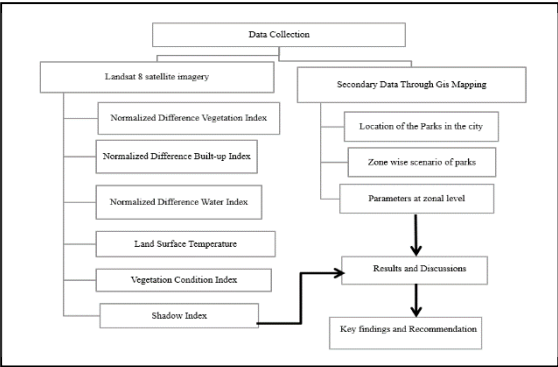


Fig 3: Methodology Flowchart

INDICES (LANSAT IMAGERY 8)	FORMULA USED	Purpose
Normalized Difference Vegetation Index (NDVI)	(Band 4 – Band 3) / (Band 4 + Band 3)	This is a prominent indicator used for identifying vegetation. A higher NDVI index score suggests more vegetation on the ground, negative NDVI index value denotes non-green areas such as barren, sea, river, and built-up areas, and a positive value represents green areas.
Normalized Difference Built-up Index(NDBI)	(Band 6 –Band 5) / (Band 6+ Band 5)	Used to identify and analyze built-up areas.
Normalized Difference Water Index (NDWI)	(Band 3 –Band 5) / (Band 3 + Band 5)	Used to delineate and monitor water bodies and assess water content in vegetation.
Vegetation Condition Index (VCI)	100 * (NDVI - NDVI min) / (NDVI max - NDVI min)	Monitoring environmental health, detecting drought and other stressors, assessing land cover changes.
Shadow Index (SI)	((1-Band 2)* (1-Band 3)* (1-Band 4))* (1.0/3.0)	Used to identify and examine shadows, which are important for precise vegetation mapping, land cover interpretation, and other geospatial analysis.

Table 1 Steps for Different Indices Used in the Study

4.	$PV = ((NDVI - NDVI_{min}) / (NDVI_{max} - NDVI_{min}))^2$	the city's surface temperature, climatic patterns, and other environmental processes.
5.	$E = 0.004 * PV + 0.986$	
6.	$LST = BT / (1 + (\lambda * BT / C2) * \ln(E))$	

Table 2 Steps for Land Surface Temperature Used in the study

IV. RESULTS AND DISCUSSIONS

The inadequacy of open spaces became increasingly evident as Bangalore transformed into a major urban center.

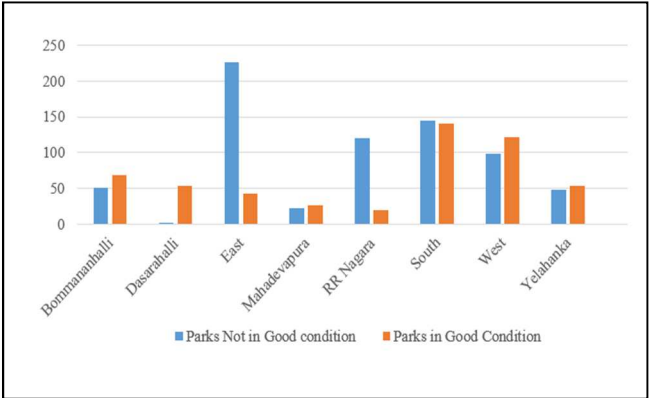


Figure 4 Status of present condition of parks Zone-wise

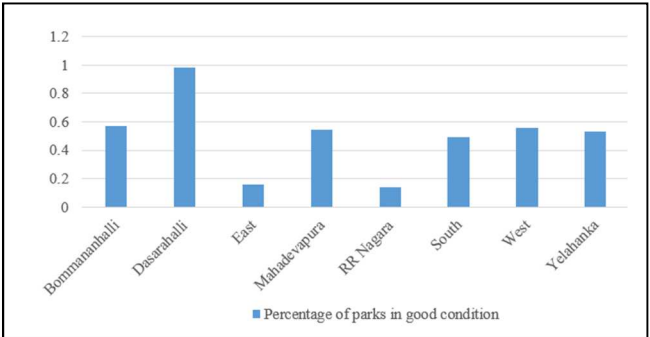


Figure 5 Percentage of parks in good condition Zone-wise

From figure 4 and 5, shows that South Zone has the highest no. of parks, followed by East, West, RR.Nagara, Bommanahalli, Yelahanka, Dasarahalli and Mahadevapura Zone. The city core has reasonable access to organized green open spaces. And if we see the highest percentage of parks in good condition are from Dasarahalli zone and least percentage of parks in good condition are from RR.Nagara. The periphery of the city has the least provisioning of organized green open spaces [15]. Area under park in the city right now is approx. 1.6 sq.m per person and according to the standards the area should be 9-10 sq.m. The scenario of parks is that 79% Parks located in inner wards, 21% Parks closed to the public, 6% parks have public toilet facilities, 54% Parks contains garbage / debris, 3% Parks have drinking water facilities, 36% parks have playing equipment facilities, 56% Parks have lighting facilities, 72% Parks have security staff,

INDICES (LANSAT IMAGERY 8)	FORMULA USED	Purpose
Normalized Difference Land Surface Temperature (LST)	1. $L\lambda = ML * Band\ 10 = AL - Qi$, where $L\lambda = 0.0003342$, $AL - Qi = 0.10000 - 0.29$ 2. $BT = k2 / \ln(K1 / L\lambda - 1) - 273.15$, where $k2 = 1321.0789$, $k1 = 774.8853$ 3. $NDVI = (Band\ 4 - Band\ 3) / (Band\ 4 + Band\ 3)$	Monitoring temperature changes and evaluating the effects of human activity on the environment are made possible by this crucial indicator of

80% Parks have jogging tracks, 77% Parks have seating areas [19].

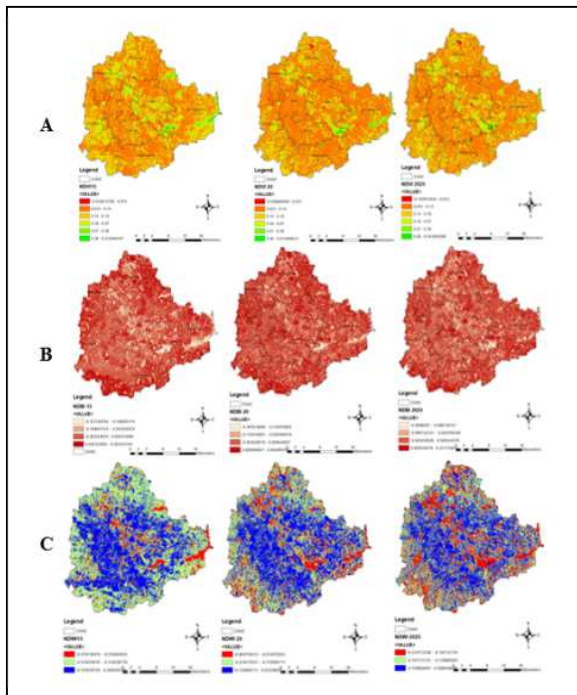


Fig 6: Maps of different indices A- NDVI (2015, 2020, 2025 respectively) B- NDBI (2015, 2020, 2025 respectively) C NDWI (2015, 2020, 2025 respectively)

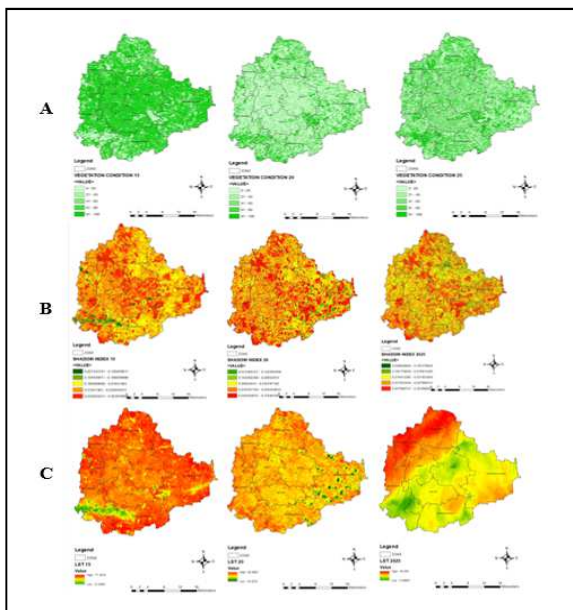


Fig 7: Maps of different indices A- VCI (2015, 2020, 2025 respectively) B- SI (2015, 2020, 2025 respectively) C- LST (2015, 2020, 2025 respectively)

From fig 6 and 7 it is clearly visible that there are lot of changes happened in the stretch of 10 years. Depletion of water bodies over 10 years has been observed the reason could be urbanization and increase in built-up areas. Similarly we can see that Land surface temperature is very

high in central part of the city may due to less or no presence of vegetation or green areas. Fig 8 shows the detail analysis of the results of each zone generated from the satellite maps.

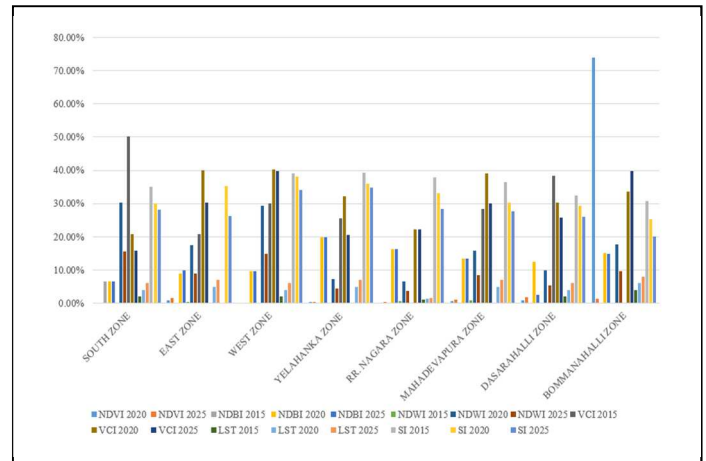


Fig 8: Graph showing the variations for each indices across zones

Fig 8 shows the variations of different indices at different years zone wise. South Zone has good amounts of parks with dense vegetation, low built-up areas, and moderate amount of water bodies, has low temperature variations and has a more amount of tree cover. Like in 2015 the vegetation cover was 0.03% and in 2020 it was 0.08 % and in 2025 it is 0.06 %. In East Zone it has parks but the number is few and are scattered all over with low vegetation and has sparse vegetation. Has high and extensive built-up areas with minimal water bodies, temperature is generally high due to less and stressed vegetation. In 2015 the vegetation percentage was 0.82% and in 2025 it is 1.63%. In west Zone parks are limited in number and in smaller size with moderate vegetation cover. Built-up has increased specially the encroachment. Here the vegetation condition is declining hence the temperature is also increasing. In 2015 it the vegetation percentage was 0.075% and in 2025 it is 0.10%. In Yelahanka Zone parks are relatively larger in size with moderate vegetation. It has moderate built-up with mixture of open spaces. It has moderate amount of water bodies which are responsible for healthy vegetation in some pockets. In 2015 the vegetation percentage was 0.19% and in 2025 it is 0.38%. In Dasarahalli Zone parks are limited and are in smaller size and sparsely distributed with low and sparse vegetation, high built-up with minimal water bodies and limited vegetation. In 2015 the vegetation percentage was 0.86% and in 2025 it is 1.73%. In Bommanahalli Zone parks are in few numbers and are scattered, and has limited recreational spaces with low and sparse vegetation. High built-up with minimal water body. Has poor and stressed vegetation with limited tree cover. In 2015 the vegetation percentage was 0.65% and in 2025 it is 1.30%. In Mahadevapura zone parks are few in number, and are scattered and has limited access due to industrial growth with sparse and moderate vegetation. Has high amount of built-up areas due to presence of industry and IIT Hubs. Has low water bodies in some parts. Vegetation condition is moderate, slightly better near the IIT campuses. In 2015 the vegetation percentage was 0.57% and in 2025 it is 1.14%. In RR.Nagara Zone there are moderate number of parks, better

distributed than Dasarahalli and Bommanahalli, but are in smaller size. Has moderate to high vegetation, has decent vegetation cover in certain residential pockets. Has moderate and a mix of built-up areas and open spaces. There are moderate amount of water body, which are responsible for healthy vegetation in some pockets. Have low temperature when compared to other zones having industry presence. Has moderate to high healthier vegetation in some pockets with relatively higher and better tree cover. In 2015 the vegetation percentage was 0.10% and in 2025 is 0.29%. If we compare overall Land surface temperature (2025) fig 7 C is very high in Northern part and some portion of central part of the city may due to less or no presence of vegetation or green areas, industrial activity, rising in built-up etc. Like in 2015 and 2020 it was around 35 °C and in 2025 it rose to 38 °C that means 3 °C rise. For NDVI fig 6 A. we can see the increase in built-up spaces especially in Mahadevapura and Dasarahalli.

For NDBI fig 6 B the north and central part of the city is a densely populated reason could be the presence of commercial area and less number of open spaces. For NDWI fig 8 C the vegetation and water bodies increased between 2015 and 2020 but in 2025 it gradually decrease the reason could be urbanization, increase in built-up areas, overall in the span of 10 years there is a gradual decrease in water bodies, vegetation may be due to change in land use, increase in built-up footprints. Overall city has changed a lot in 10 years in respect to environment changes, physical changes which includes population, population density, land use patterns and growth of the city in peripheral parts. All this could be the reason of drastic loss of vegetation and loss of open spaces including parks and gardens.

V. CONCLUSION

The research had used geospatial methods, such as remote sensing and GIS, to examine urban parks and open spaces in Bangalore, highlighting their crucial role in data gathering and thorough assessments of green space distribution, air quality, and microclimates. By employing indices like NDVI (Normalized Difference Vegetation Index), LST (Land Surface Temperature), and VCI (Vegetation Condition Index), the study underscores environmental shifts over the past decade, including the reduction of water bodies, increased urbanization leading to more built-up areas, and rising land surface temperatures in regions with limited vegetation. Analysis by zone uncovers disparities in park distribution and conditions: the South Zone boasts dense vegetation and better park access, whereas zones like East, West, Bommanahalli, Dasarahalli, and Mahadevapura struggle with limited green spaces, sparse vegetation, and high built-up areas. The government should prioritize

interventions in RR Nagar Zone, which has the lowest percentage of parks in good condition, and in peripheral zones like Dasarahalli and Mahadevapura to improve the availability and quality of open spaces, ensuring equitable access and enhancing the overall quality of life for residents and in central zone of the city for improving vegetation cover and reducing land surface temperature for the open spaces located there.

Future research could tackle these challenges by exploring the specific causes behind environmental changes like water body depletion and increased urbanization. Additionally, studies should evaluate the impact of urban planning policies on open space preservation while examining socio-economic factors influencing park accessibility and usage. Strategies for enhancing community involvement in park planning and maintenance are crucial for equitable distribution of green spaces across diverse communities. This comprehensive approach could significantly enhance Bangalore's liveability by ensuring sustainable development while preserving its natural resources amid rapid urbanization.

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